Revision log for "Conjectural permutation decoding of some AG codes"

Corrected and expanded version, Draft 2

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November 12, 2026

Scope

This log belongs to the corrected and expanded version of the paper. It is kept in the folder `perm-decoding` of the repository

`https://github.com/wdjoyner/qrcodes-hyperelliptic`.

The files involved are these.

Name used here	File
2005 note	`perm_decoding3.tex`, the source of D. Joyner's note in *ACM SIGSAM Bulletin* 39(1) (2005), 26–32
Draft 0	`perm_decoding_step1_mathematical_revision.tex`, by D. Joyner and ChatGPT
Intermediate draft	`perm-decoding_revision-2026.tex` as committed on November 9, 2026 (commit `188d526`). It is no longer in the folder and can be recovered from the git history.
Draft 1	`paper/perm_decoding_step2_revision.tex` (November 9, 2026)
Draft 2	`perm-decoding_revision-2026.tex` (November 12, 2026)

Draft 2 was first committed in place of Draft 1 (commit `bae43d8`). At D. Joyner's request it was then moved to `perm-decoding_revision-2026.tex`, where it replaced the intermediate draft, and Draft 1 was restored unchanged to its own file. The draft number appears on the title page of Draft 2. In the history of the folder and in the file names, Draft 0 is called "Step 1" and Draft 1 is called "Step 2".

How this log was made. No revision log was kept before Draft 2. Section 1 is a reconstruction. Claude wrote it on November 12, 2026 by comparing the files listed above and by reading the Acknowledgments and the `README.md` of the folder. The referee report on Draft 0 and ChatGPT's response to it are not in the repository and were not consulted. Where Section 1 says who found something, the source is the Acknowledgments of the paper. Section 2 is a direct record of the changes made in Draft 2.

Numbering. The 2005 note numbers its lemmas, propositions, examples and conjectures with one counter (Lemma 1 to Conjecture 7), and its remark separately. Unless stated otherwise, all other numbers in this log are Draft 2 numbers. Section 2.5 gives the concordance between Draft 1 and Draft 2.

1 Draft 1: from the 2005 note to the corrected and expanded version (reconstructed)

1.1 Stages

1. *Draft 0.* ChatGPT revised the mathematics of the 2005 note. The result had the title "Permutation decoding of codes from $y^2 = x^p - x$: corrected results, counterexamples, and partial decoding" and the authors D. Joyner and ChatGPT.

2. *Referee report.* Claude refereed Draft 0.

3. *Intermediate draft.* ChatGPT's response to the report and the referee's corrections were combined. Claude was added as third author.

4. *Draft 1.* The original title was restored, with the subtitle "Corrected and expanded version", and the computations for $p = 23$ were added.

1.2 Format and front matter

1. The paper was reset in 12pt Times with one-inch margins, numbered by section, with `\label` and `\ref` throughout.

2. The abstract was rewritten to state the corrected results.

3. A table of contents, a background section, a literature review (before and after 2007), a summary, a list of open problems, a glossary and two appendices were added. The appendices hold the GUAVA session of the 2005 note and a verification script.

4. The bibliography grew from 30 entries (2005 note) to 32 (Draft 0) to 59 (Draft 1).

5. The GUAVA session of the 2005 note was kept, with two typos corrected: `Length(P)` became `Length(Pts)`, and the closing `if;` became `fi;`.

1.3 Statements of the 2005 note that were corrected

Each of these corrections is flagged in the body of the paper at the place given.

1. *Point labels.* The 2005 note takes $P_1 = (1 : 0 : 1)$ and says its stabilizer is generated by $(x, y) \mapsto (x, -y)$, (a^2x, ay) and $(x + 1, y)$. The translation does not fix that point. These maps generate the stabilizer of the point at infinity. The paper reads P_1 as P_∞ and says so (Remark 4.1).

2. *Order of the stabilizer over $\mathbb{F}_p$.* The 2005 note gives $2p(p-1)$. The correct order is $p(p-1)$. The value $2p(p-1)$ is the order of the stabilizer over $\mathbb{F}_{p^2}$ (Section 4.1, after equation (4.5)).

3. *Action of the hyperelliptic involution.* The proof of Lemma 2 of the 2005 note says that ι acts trivially on the monomial basis of $L(D)$. It acts by -1 on the monomials that involve y (proof of Proposition 4.5).

4. *Lifting of Möbius transformations.* The paper states and proves the criterion that a class in $\mathrm{PGL}_2(p)$ lifts to an $\mathbb{F}_p$-automorphism exactly when its determinant is a square (Section 4.1). According to the Acknowledgments, the referee found an error in the lift criterion of Draft 0.

5. *Ample divisors and separation of points.* The 2005 note says that an ample divisor D has $L(D)$ separating points. On a curve this is false in general (Remark 5.2).

6. *Kernels.* The description of the kernel of the map from the automorphism group to the permutation group of the code was corrected (Proposition 5.1).

7. *Permutation representations.* Proposition 4 of the 2005 note says that, under degree hypotheses, the representation on $L(D)$ is equivalent to one by permutation matrices. This is false. Only the weaker statement holds that $L(D)$ is isomorphic to a submodule of a permutation module (Proposition 5.1(3), Example 5.3).

8. *Group-action convention.* The convention $g \cdot f = f \circ g^{-1}$ was fixed and used throughout (equation (3.1)).

1.4 The two conjectures

1. Conjecture 6 of the 2005 note, called (C1) in the paper, is false as a universal statement. At $p = 37$ and $k = 17$ no information set works (Theorem 7.8). The counterexample is already in Draft 0.

2. Conjecture 7 of the 2005 note, called (C2), is false as a universal statement. Draft 0 gave one counterexample, at $p = 7$ (its Theorem on the $[84, 42, \geq 40]$ code). Draft 1 replaced it by a general obstruction for regular groups (Theorem 8.1) and an infinite family (Corollary 8.2). The case $p = 7$ became Example 8.3.

1.5 Changes between Draft 0 and Draft 1

1. The original title was restored, and Claude was added as third author.

2. The Background and Literature review sections, the Glossary and Appendix B were added.

3. Proposition 7.5 (exhaustive computations for $p \leq 23$), Remark 7.6 and Example 7.7 (failure at $p = 17$ despite the counting bound) were added. The intermediate draft covered $p \leq 19$, and Draft 1 extended this to $p = 23$.

4. Theorem 8.1, Corollary 8.2 and the table after it were added, as described above. According to the Acknowledgments, Claude proposed the logarithmic obstruction, and ChatGPT found the choice $m = n/2 + g - 1$ and the information set of Example 7.7.

5. Problems 10.2 and 10.4 and the references to the recent literature on permutation decoding of AG codes were added.

6. The corrections 1, 4, 6 and 8 of Section 1.3 were made at this stage, according to the Acknowledgments.

2 Draft 2: revision for the style guide (November 12, 2026)

2.1 Scope

Claude (Claude Fable 5.2) brought the paper into line with D. Joyner's style guide, following his instructions of November 12, 2026. The guide is the file `joyner-math-paper-style-guide.md` at the top level of the repository (Draft 2 of November 11, 2026). Those instructions forbade mathematical changes. No statement, proof, hypothesis or status label was altered. All new mathematical text was confined to what the guide requires, and it is listed in Section 2.4 so that it can be refereed. Semicolons in existing text were left in place. Rule 3.8 (no semicolons) was applied to new text only. The guide has no paper-specific rule (its Section 9) for this paper.

Before the changes, the paper compiled with four errors, all at the title of Example 7.4. The optional argument of that example contained unprotected square brackets, so the title was cut short and the first sentence of the example was set in math mode. There were also 16 hyperref warnings, caused by math in five section titles, one overfull box (the display (4.4), by 53pt) and two underfull boxes in the bibliography. After the changes, the paper compiles with no errors, no warnings, no undefined references and no overfull boxes. One underfull box remains, in the bibliography entry `Go03`.

2.2 Defaults used for the rules marked [DECIDE]

1. Document class: the existing class, `article`, was kept.

2. Times package: `newtxtext` and `newtxmath`.

3. Glossary placement: notation near the beginning (Subsection 3.7), and the glossary at the end, just before the Acknowledgments.

4. AI systems on the author line: the instructions did not mention the author line, so it was left as it was (D. Joyner, ChatGPT, Claude). The Acknowledgments also credit both systems and point to this log.

2.3 Changes

Typesetting.

1. The class and `geometry` options now name letter paper. The conditional fallback to `mathptmx` was removed, and `newtxtext` and `newtxmath` are loaded directly.

2. The title of Example 7.4 is now enclosed in braces inside the optional argument.

3. The display (4.4) was too wide by 53pt. It is now set on two lines. Its content and its number are unchanged.

4. The five section and subsection titles that contain math now use `\texorpdfstring`, which removes the hyperref warnings.

5. In Subsection 2.2 the longer key `CLLS25` made one line too wide. The citation was moved four words later in the same sentence, which now reads "... and Santos gave an elementary proof [CLLS25] that ... ".

Title page, abstract, Acknowledgments.

6. The subtitle became "Corrected and expanded version. Draft 2" (rule 6.2). The title is unchanged.

7. The date line became "Original draft: 2005. Corrected and expanded draft: 2026." (rule 6.2).

8. A `\thanks` note on the title identifies the 2005 note and gives the date of Draft 2 (rule 6.3).

9. The Naval Academy address was removed from the first author's footnote, which now reads "Original author. `wdjoyner@gmail.com`" (rule 5.1).

10. The first sentence of the abstract, "This AI-aided paper revisits ... ", was replaced by a disclosure sentence that names both AI systems and refers to the Acknowledgments, followed by "The paper revisits ... " (rule 5.2). The rest of the abstract is unchanged.

11. The Acknowledgments gained two paragraphs. The first describes Claude's role in Draft 2 and cites the check script by commit (rules 5.3 and 8.4). The second says that AI review is not an independent certification and points to this log (rules 5.4 and 5.5).

Citation keys.

12. Sixteen keys and their labels were changed to the pattern of rule 1.7, and every `\cite` was updated.

Old	New	Old	New	Old	New	Old	New
CLLS	CLLS25	HvLP	HvLP98	JT	JT02	Sti	St93
G	Go03	HP	HP03	KW	KW90	TV	TV91
Gu	Gu04	JK2	JK03	L	Le82	W	We98
H	Ha77	JKT	JKT08	S	Sc90	repo	Jo26

The label of `Du87` lost its umlaut. The years in `Go03`, `JK03`, `JT02` and `JKT08` are taken from the arXiv numbers in the existing entries. No bibliographic data was changed.

13. The entry `Jo26` (the repository) was moved next to `Jo05`. The widest label of the bibliography is now `JLJHH89`.

Section order.

14. The Literature review now comes before the Background (rule 2). They were Sections 3 and 2 and are now Sections 2 and 3.

15. The former Section 10 (Open problems) and Section 11 (Summary) were merged into one Section 10, "Summary of results and open problems", with the summary first (rule 2.7). The label `sec:problems` now points to the merged section. The problems keep the numbers 10.1 to 10.4.

16. The Acknowledgments were moved from before the appendices to after the Glossary, just before the references (rule 2.9).

17. The "Organization" paragraph of the Introduction was updated to match the new order.

2.4 New text to be refereed

Rules 2.4, 2.5, 3.2, 3.3 and 3.4 required the following additions. Every number in N1 to N10 is checked, by exhaustive and exact computation, by the script `check_examples_draft2.py` in the directory `scripts` (commit `9e1e1ea`).

N1. Example 1.1 (`ex:intro-p7`). The $[7, 3, 5]$ code as an instance of the setup. It restates facts of Example 7.4.

N2. Example 3.1 (`ex:bg-codes`). Information set, check positions and the parity-check matrix for the generator matrix of Example 7.4. The parity-check matrix is new.

N3. Example 3.2 (`ex:rs7`). Two codewords of $\mathrm{RS}(\mathbb{F}_7, 3)$ and one affine map.

N4. Example 3.3 (`ex:bg-p1`). $\mathrm{RS}(\mathbb{F}_7, 3)$ as an AG code on the projective line.

N5. Example 3.4 (`ex:bg-hyper`). Weierstrass points and point counts of $y^2 = x^7 - x$ over $\mathbb{F}_7$ and $\mathbb{F}_{49}$.

N6. Example 3.5 (`ex:bg-groups`). The action of $\mathrm{AGL}(1, 7)$ on $\mathbb{F}_7$. It includes a new small claim: the sum-zero submodule of $\mathbb{F}_7^7$ is not a permutation module for $\mathrm{AGL}(1, 7)$. The argument is the one used in Example 5.3.

N7. Example 3.6 (`ex:bg-pgl`). Orders of $\mathrm{PGL}_2(7)$, $\mathrm{PSL}_2(7)$ and the Borel subgroup, and two maps that are and are not in $\mathrm{PSL}_2(7)$.

N8. Example 4.8 (`ex:curve-p7`). The case $p = 7$ of Section 4: a basis of $L(7P_\infty)$, the orders of G_7, H_7, G_{49} and H, the orbit sizes, and $\ell(D_2) = 82$. It also serves as the example for Definition 4.3 (large group).

N9. Example 6.5 (`ex:pd-translations`). It serves as the example for Definition 6.2. It contains two new small claims. The seven translations form a 2-PD-set for $\mathrm{RS}(\mathbb{F}_7, 3)$ relative to $\mathcal{I} = \{0, 1, 2\}$. No set of three permutations is a 2-PD-set for that code.

N10. Example 9.1 (`ex:cost`). Operation counts for $p = 7$ and for $\mathrm{RS}(\mathbb{F}_{23}, 11)$.

N11. **Table of notation** (Subsection 3.7), with an orienting paragraph (rule 2.5). Each row points to the place where the symbol is introduced.

N12. **Orienting paragraphs** (rule 3.2) at the start of Sections 1, 2, 4, 5, 6, 7, 8, 9 and 10, of Subsections 2.1, 2.2, 3.1 to 3.6, 4.1, 4.2, 5.1, 7.1, 7.2, 10.1 and 10.2, and of the Glossary. Two sentences were added to the existing opening paragraph of Section 3. The paragraph of Subsection 10.1 makes a statement about status: the permutation group in item 1 of the summary is cited from the literature, item 3 rests in part on the computer search of Proposition 7.5, and everything else is proved in the paper. A referee should confirm that this describes the existing results correctly.

N13. Seven new labels for the subsections of Section 3 (`subsec:bg-codes` and so on, and `subsec:notation`).

2.5 Concordance of numbers

Every `\label` of Draft 1 was kept. The table was generated by comparing the `.aux` files of the two drafts. All new examples were placed either in sections that had no numbered statements or at the end of a section, so no theorem, proposition, lemma, corollary, remark, example or problem of Draft 1 changed its number.

Draft 1	Draft 2	Label or title
Section 2	Section 3	`sec:background`
Subsections 2.1–2.6	Subsections 3.1–3.6	(Background)
Equation (2.1)	Equation (3.1)	`eq:convention`
Section 3	Section 2	`sec:lit`
Subsections 3.1, 3.2	Subsections 2.1, 2.2	(Literature review)
Section 10	Subsection 10.2	Open problems (`sec:problems` is now Section 10)
Section 11	Subsection 10.1	Summary

New numbers in Draft 2: Examples 1.1, 3.1–3.6, 4.8, 6.5 and 9.1, and Subsection 3.7.

2.6 Rules not applied, and why

D. Joyner approved items 1 to 5 before the changes were made.

1. Rule 6.4 for the corrections of the 2005 note in the body (Remark 4.1, the note on the stabilizer order in Section 4.1, the proof of Proposition 4.5, Remark 5.2, Subsection 5.1, the paragraph "Relation to the original note"). They are mathematical statements about a published paper, and removing them would change the mathematics.

2. Rule 3.3 in Section 2 (Literature review), Section 10 (Summary of results and open problems), the appendices and the Glossary. An example there would add nothing.

3. Rule 1.7 for the key `GUAVA`, which is software cited by name.

4. Rule 8.3 as a draft number in the file name. The draft number is on the title page only.

5. Rule 1.6, second sentence (unnumbered subsections). Rule 3.8 in existing text. Rules 3.6 and 3.7, where no violation was looked for. Rule 6.6, since no status label was added.

6. Rule 2.3 asks that the literature review be split at the date of the original paper, which is 2005. The existing split is at 2007. Moving it would mean rewriting the paragraphs on work of 2006 and 2007, so it was left for D. Joyner to decide.

7. Rule 1.5 for citations typed by hand. None was found.

8. Rules 8.5 and 8.6 concern the repository. It has no license file. The copyright position of this corrected version relative to the published 2005 note is not recorded. Neither point was acted on.

2.7 Not verified

1. No cited source was consulted in Draft 2 (rule 4.3). See Section 4.

2. The computations in the body were rerun in Draft 2 (rule 4.4) and confirmed. The program `pd_affine.py` was rerun for every prime $5 \le p \le 23$. Its output agrees with the stored files in `results` apart from the timings, and the table of Proposition 7.5 agrees with that output entry by entry. The order 42 of the permutation group of the $[7, 3, 5]$ code was confirmed by a search of all of S_7 in the check script. The other numerical statements of the body (Example 7.7, Theorem 7.8, Remark 7.9, the table after Corollary 8.2 and Example 8.3) were rechecked with the new script `check_body_numbers.py`, and all were confirmed. No hand proof in the body was rechecked.

3. The new text N1 to N13 was checked by its author and by the check script, not by an independent referee.

4. Section 1 is a reconstruction, as explained under Scope.

5. The versions of ChatGPT and Claude that did the work before Draft 2 are not recorded in the repository.

3 AI usage

1. *ChatGPT (OpenAI): first reviser.* It prepared Draft 0 and responded to the referee report. The Acknowledgments of the paper list its contributions.

2. *Claude (Anthropic): referee, co-author and reviser.* It refereed Draft 0, carried out the exhaustive computations of Proposition 7.5, wrote the scripts in `scripts/`, and prepared Draft 1, including the Background, the Literature review and the Glossary. The Acknowledgments list its contributions.

3. *Claude (Claude Fable 5.2): style reviser, Draft 2.* It made the changes in Section 2, wrote the new text N1 to N13 and the check script for it, and wrote this log.

D. Joyner directed the work and is responsible for the final content. AI review is not an independent certification of the results.

4 Sources not consulted

Rule 4.3 asks that every citation be checked against its source. In Draft 2 no cited source was consulted. The Acknowledgments of Draft 1 say that Claude verified the bibliography at that stage, but the repository holds no record of which sources were read. In particular, the following precise citations were not checked in Draft 2.

1. From Stichtenoth [St93]: Theorems I.5.17, II.2.2 and V.2.3, Proposition VI.4.1 and §VII.3.
2. From Hartshorne [Ha77]: Corollary IV.3.2.
3. From MacWilliams and Sloane [MS77]: Chapters 11 and 16.
4. The theorem of Knörr and Willems [KW90] used in Theorem 7.1.

5 Original and new results

Rule 6.5 asks that original results be separated from results first established in the revision. Draft 2 adds no new results apart from the small claims in N6 and N9.

2005 note	Draft 2
Definition of a large group	Definition 4.3
Lemma 1 (large groups ramify)	Lemma 4.4
Monomial basis of $L(mP)$, quoted from Stichtenoth	Lemma 4.2 (with proof)
Lemma 2 (semisimplification)	Proposition 4.5 (statement on ι in the proof corrected)
Proposition 3 (orbits over $\mathbb{F}_{p^2}$, proof omitted)	Proposition 4.6 (with proof)
Kernel and separation of points (Section 3.1)	Proposition 5.1, Remark 5.2 (corrected)
Proposition 4 (permutation matrices)	false, replaced by Proposition 5.1(3) and Example 5.3
Syndrome lemma, stated without proof	Lemma 6.1 (with proof)
Example 5 (the $[7, 3, 5]$ code)	Example 7.4
Conjecture 6	(C1), refuted by Theorem 7.8
Conjecture 7	(C2), refuted by Corollary 8.2
—	New in the revision: Theorem 6.3 (counting bound, after Gordon), Theorem 7.1 (Reed–Solomon identification), Theorem 7.3 (two errors), Proposition 7.5, Remark 7.6 and Example 7.7 (computations for $p \le 23$), Theorem 7.8, Theorem 8.1, Corollary 8.2, Example 8.3, Example 5.3, Problems 10.1–10.4

References

[Ha77] R. Hartshorne, *Algebraic Geometry*, Springer, 1977.

[KW90] R. Knörr and W. Willems, The automorphism groups of generalized Reed–Muller codes, *Astérisque* 181–182 (1990), 195–207.

[MS77] F. J. MacWilliams and N. J. A. Sloane, *The Theory of Error-Correcting Codes*, North-Holland, 1977.

[St93] H. Stichtenoth, *Algebraic Function Fields and Codes*, Springer, 1993 (2nd ed. 2009).